

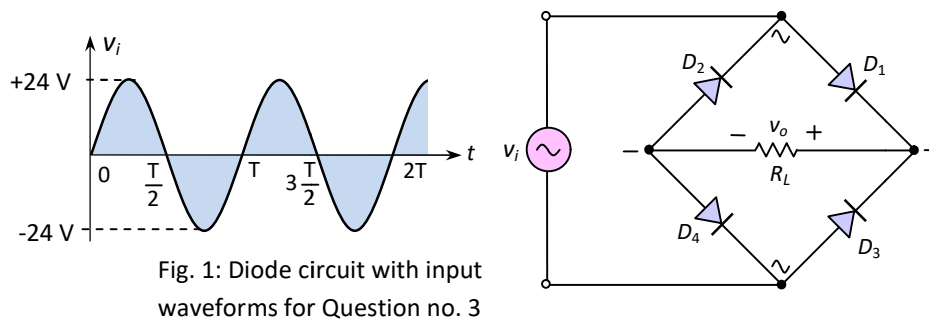


EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Quiz I, Semester: Fall 2023

Course: CSE 251 –Electronic Circuits, Section-1
Instructor: Dr. Md. Habibur Rahman, Adjunct Professor, CSE Department
Full Marks: 15 (Will be converted into 10)
Time: 50 Minutes

Note: Answer ALL of the questions. Course outcomes (CO), cognitive levels and marks of each question are mentioned at the right margin.

- Write the Shockley's equation for PN junction diode. [CO1, Mark 3.5]
 Suppose a PN junction has the following values:
 $n = 1$, $T = 27^\circ$, $I_S = 10$ nA. **Calculate** the diode current (I_D) if it is forward biased by $V_D = 0.75$ V.
- As depicted in Fig.1, an AC voltage is applied to the diode circuit. Considering Si diodes, **sketch** the output waveforms and **calculate** the RMS value of the output. [CO2, Mark: 3.5]



- Calculate I_R , I_Z , I_L for the voltage regulator circuit given in Fig.5 [CO2, EP1, Mark: 4]

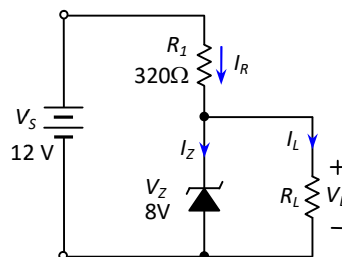


Fig. 5: Diode circuit for Question no. 6

- Why is the name transistor? Draw the output characteristics of a BJT in CEC and describe its three regions. CO1 Mark: 4